MIL-A-46091B
22 July 1981
SUPERSEDING
MIL-A-46091A
4 March 1965

MILITARY SPECIFICATION

ADHESIVE, BRAKE LINING TO METAL

This specification is approved for use by Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of thermosetting, one-component, adhesive for use in bonding brake linings to steel and aluminum (see 6.1).

1.2 Classification. The adhesive shall be of the following forms, as specified by the procuring activity (see 6.2).

Form I - Liquid
Form II - Film

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

HH-L-361	- Lining, Friction and Lining Material, Friction (Brake) Automotive Material
QQ-A-225/8	- Aluminum Alloy Bar, Rod, Wire and Special Shapes; Rolled, Drawn, or Cold Finished, 6061
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet
QQ-S-698	- Steel, Sheet and Strip, Low Carbon
VV-B-680	- Brake Fluid, Automotive
PPP-B-636	- Box, Shipping, Fiberboard
PPP-B-1055	- Barrier Material, Waterproof, Flexible
PPP-D-723	- Drums, Fiber
PPP-P-1892	- Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking of
PPP-T-60	- Tape, Pressure, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-LS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation, Packing, Methods of
MIL-B-46176 - Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and Submission of

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connections with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A331 - Steel Bars, Alloy, Cold Finished

SOCIETY OF AUTOMOTIVE ENGINEERS

Recommended Practice

SAE J840 - Test Procedures for Brake Shoe and Lining Adhesives and Bonds

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 2 Pennsylvania Ave., New York, NY 10001.)

UNIFORM FREIGHT CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules and Container Specification

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, IL.)

CODE OF FEDERAL REGULATIONS

49-CFR-0-190 - Department of Transportation Rules and Regulations for the Transportation of Hazardous Materials.

(Copies of these regulations may be purchased from the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. The adhesive shall be one component, heat curing material. There shall be no restrictions on the chemical type or physical nature of materials used in the adhesive except those imposed by other technical requirements of this specification.

3.1.1 Liquid form. Liquid adhesive shall be of a consistency suitable for application (see 3.4).

3.1.2 Film form. Film adhesive shall consist of unsupported film, protected by a liner. The film thickness shall be within the range of 0.2 to 0.3 ml (0.008 to 0.012 in.). Unless otherwise specified, the adhesive shall be furnished in rolls of the length and width normally supplied by the manufacturer (see 6.2).

3.1.2 Solvent. If a solvent is required as a separate thinner or activator, it shall be furnished by the adhesive manufacturer (see 6.2). When solvents or adhesives are of a flammable nature, the necessary precautions specified in 5.4 shall be observed and proper caution labels applied to the containers and packages.

3.2 Working characteristics.

3.2.1 Adhesive preparation. Adhesives shall be thinned, heated or otherwise prepared for use per manufacturer's instructions (see 3.4).

3.2.2 Application. Adhesives shall be capable of being readily applied to thoroughly clean surfaces of aluminum alloys, steel, and to all types, styles and classes of brake lining HH-L-361, except Type I, Style 2. Application shall be in accordance with the manufacturer's instructions (see 3.4).

3.2.3 Curing conditions. The heat, pressure and time required for curing shall be in accordance with the manufacturer's procedure (see 3.4).

3.3 Physical properties. Physical properties of the test specimens prepared and tested as specified in 4.5 shall conform to the requirements listed in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

Table I. Requirements

Property	Value	Test Paragraph
Disk shear strength:		
Tested @ ambient temperature:		
Steel to steel	6.89 MPa min. (1000 psi)	4.5.1
Aluminum to aluminum	6.89 MPa min. (1000 psi)	4.5.1
Tested @ 204°C (400°F):		
Steel to steel	6.89 MPa min. (1000 psi)	4.5.1
Aluminum to aluminum	6.89 MPa min. (1000 psi)	4.5.1
Axial shear strength:		
Tested @ ambient temperature:		
Brake lining to steel	Pass[1]	4.5.2
Brake lining to aluminum	Pass[1]	4.5.2
After brake fluid immersions (test each of 3 fluids):		
Brake lining to steel	Pass[1]	4.5.3
Brake lining to aluminum	Pass[1]	4.5.3
After water immersion:		
Brake lining to steel	Pass[1]	4.5.4
Brake lining to aluminum	Pass[1]	4.5.4
Tested @ 204°C (400°F):		
Brake lining to steel	Pass[1]	4.5.5
Brake lining to aluminum	Pass[1]	4.5.5

[1]The bond shall be considered satisfactory provided the following conditions are met:

a) The shear load shall be adequate as mutually agreed upon by the supplier and purchaser.

b) Not less than 70% of the intended bond area on the shoe table or rim shall be covered with lining residue of perceptible thickness (see SAE J810; fig. 8 No. 6 or 7 fracture).

c) The adhesive is completely polymerized. This can be determined by a hot shear test or by wetting a small area of the adhesive on the shoe table or rim with a suitable solvent. If the adhesive does not become tacky to the touch or cannot be dissolved with the solvent, the adhesive is considered completely polymerized.

d) No more than 30% of the total intended bond area nor more than 10% of the total intended bond area within 1.3cm (1/2 in.) of each end of the lining shows bare metal (see SAE J840; fig. 8, No. 1 fracture).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

3.4 Instruction sheet. Unless otherwise specified, the manufacturer shall provide three dated and titled instruction sheets with each shipment of adhesive (see 5.5 and 6.2) outlining instructions for its use on brake lining, aluminum alloy and steel parts (see 3.2.2). The instruction sheet shall include the following information:

a) The general chemical type of the base resin used in the adhesive, such as phenolic-neoprene, epoxy or vinyl-phenolic.

b) Instructions on the use of a thinner or solvent with the adhesive, including minimum, optimum and maximum amounts to be used.

c) Detailed instruction for any necessary procedures prior to use of the adhesive, such as stirring, thinning or heating (for viscosity control or hot coating). The temperature controls necessary to maintain the adhesive in a form suitable for application and the maximum pot life of the adhesive when maintained at the temperatures specified for application.

d) Complete processes and treatment for preparing the faying surfaces prior to their bonding with the adhesive. These shall include a parts-by-weight listing of all ingredients of the metal cleaning media and etch solutions, full details of mixing and using procedures, temperatures, times, rinsing media and drying requirements; also maximum allowable storage time, necessary temperature-humidity controls, and protective measures necessary during storage before application of adhesive.

e) Application instructions including spread method, number of coats, spread rate, ranges for application temperature and relative humidity, and acceptable glue-line (bond) thickness range after drying (see also 3.4 (i)).

f) Drying time between coats and after last coat. If a force dry is required, the time and temperature shall be stated.

g) Maximum allowable open assembly time and temperature-humidity controls and protective measures necessary during storage for adhesive-coated metal prior to assembly.

h) Maximum allowable closed assembly time and temperature-humidity controls and protective measures necessary during storage for adhesive-coated and assembled parts prior to cure.

i) Typical time, temperature and pressure for each segment of the complete curing cycle giving maximum and minimum limits for each condition.

j) Maximum storage life of the unopened containers at the recommended temperature. When the recommended temperature is other than $24^{o} \pm 3^{o}C$ ($75^{o} \pm 5^{o}F$) the storage life at this temperature shall be included.

k) Necessary safety precautions to be observed throughout all operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

l) Any other pertinent information relative to the use and storage of the adhesive.

3.5 Identification markings. The manufacturer shall designate each form of adhesive by a trade name and code number which shall be used to identify the adhesive.

3.6 Workmanship. Liquid adhesives shall be readily stirred to a smooth, uniform consistency throughout and shall contain no lumps, grit or other foreign matter. Film adhesives shall be free of holes, folds, wrinkles and foreign matter.

3.7 Toxicity. When used for its intended purpose, the adhesive shall have no adverse effect on the health of personnel (see 6.3).

3.8 Material safety data sheet. When specified (see 6.2) a material safety data sheet shall be prepared and submitted in accordance with FED-STD-313.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Lot. A lot shall consist of adhesive manufactured by one manufacturer, subjected to the same manufacturing and processing conditions, of one form only and offered for delivery at one time.

4.3 Sampling

4.3.1 For examination. A random sample of filled containers of liquid adhesive or complete rolls of film adhesive shall be taken in accordance with Level II of MIL-STD-105.

4.3.2 For test. A sample of the liquid or film adhesive sufficient to determine conformance of the material to the requirements of this specification shall be taken at random from each lot. Failure of the sample to meet the requirements and tests shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

4.4 Examination. Sample units selected in accordance with 4.3.1 shall be examined for the defects shown in Table II.

Table II. Classification of defects

Categories	Defects	Method of examination
Major		
	Material (AQL 1.5%)	
101	Not one component (see 3.1)	Visual
102	Not uniform upon stirring (see 3.6)	Visual
103	Presence of lumps, grit or foreign matter (applicable to liquid adhesive see 3.6)	Visual
104	Presence of holes, folds, wrinkles or foreign matter (applicable to film adhesive see 3.6)	Visual
105	Improper thickness (see 3.1.2)	Gage[1]
106	Liner not present (see 3.1.2)	Visual
	Preparation for delivery (AQL 4.0%)	
107	Improper size container (see 5.1.1.1)	Visual
108	Improper width and length (see 3.1.2)	Gage[1]
109	Improper packaging (see 5.1)	Visual
110	Improper packing (see 5.2)	Visual
111	Improper marking (see 5.4)	Visual
112	Incomplete or no instruction sheet (see 3.4)	Visual
113	Improper packaging of instruction sheet (see 5.5)	Visual

[1]Gage approved by procuring activity.

4.5 Tests

4.5.1 Disk shear test. The disk shear test shall be conducted in accordance with the method of SAE J840B. Tests shall be conducted on five bonded steel to steel (ASTM A331, Grade 1020) and five bonded aluminum to aluminum (QQ-A-225/8) specimens. The method of specimen preparation shall be in accordance with SAE J840B, except that all faying surfaces shall be cleaned, the adhesive prepared and applied and the test specimen assembly bonded in accordance with the adhesive manufacturer's recommendations. The average shear strength values shall be reported to determine conformance to the requirements of Table I.

4.5.2 Axial shear test. Axial shear strength test shall be conducted in accordance with SAE J840B. Five specimens shall be prepared from brake lining (Type II of HH-L-361 bonded to steel QQ-S-698, Grade 1020) and five to aluminum (QQ-A-250/11).

4.5.3 Brake fluid resistance. Axial shear strength shall be determined in accordance with 4.5.2, except that prior to testing, the specimen shall be immersed for 24 $\pm$ 1/4 hours in nonpetroleum base brake fluids conforming to the requirements of VV-B-680 and MIL-B-46176 in accordance with procedures outlined in SAE J840B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

4.5.4 Water resistance. Axial shear strength shall be determined in accordance with 4.5.2, except that prior to testing, the specimen shall be immersed for 24 ± 1/4 hours in tap water in accordance with the procedures outlined in SAE J840B.

4.5.5 Resistance to elevated temperature. Axial shear strength tests shall be determined in accordance with 4.5.2, except that SAE J840B, and the test temperature shall be 205 ± 3°C (400 ± 5°F).

5. PACKAGING

5.1 Packaging of liquid and film adhesives.

5.1.1 Liquid adhesive.

5.1.1.1 Level A, B and C. Packaging shall be in accordance with PPP-P-1892. Level of packaging and type, class, shape and size of containers required shall be as specified by the procuring activity (see 6.2). When a solvent is required as a thinner or activator, the adhesive and the required amount of solvent shall be secured together so that they will not become separated during shipment or storage.

5.1.2 Film adhesive.

5.1.2.1 Level A. Film adhesive shall be wrapped with at least one layer of draft waterproof wrapping paper, conforming to PP-B-1055, Class C-1, and tightly sealed with tape conforming to PPP-T-60.

5.1.2.2 Level C. Film adhesive shall be packaged in such a manner that will insure against deterioration and damage during shipment and handling from the supply source to the first receiving activity.

5.2 Packing of liquid and film adhesives.

5.2.1. Liquid adhesive.

5.2.1.1 Level A, B and C. Packing shall be in accordance with PPP-P-1892. Level of packing required shall be as specified by the procuring activity (see 6.2).

5.2.2 Film adhesive.

5.2.2.1 Level A. Packaged film adhesive of 5.1.2.1 shall be packed in boxes conforming to PPP-B-636 Class Weather-resistant, V3C or V3S, or fiber drums (PPP-D-723, Type II or III, Grade A). Closures shall be in accordance with the applicable specification.

5.2.2.2 Level B. Packaged film adhesive for Level B shall be the same as Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

5.2.2.3 Level C. The adhesive shall be packed in a manner to insure carrier acceptable and safe delivery to destination at lowest rates. Shipping containers shall comply with the Uniform Freight Classification Rules and Regulations applicable to the mode of transportation.

5.3 Department of Transportation Regulations. In addition to the requirements specified in 5.1 and 5.2, the adhesive shall be packaged and packed in accordance with the applicable requirements of the rules and regulations for the transportation of hazardous materials of the Department of Transportation (see 6.5).

5.4 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129, and with applicable requirements of the rules and regulations for the transportation of hazardous materials, of the Department of Transportation.

5.5 Instruction sheet. Unless otherwise specified in the contract or purchase order, the instruction sheet specified in 3.4 shall be packaged in accordance with method IC-3 of MIL-P-116. Packages shall be clearly marked "Instruction Sheet".

6. NOTES

6.1 Intended use. These adhesives are intended for use in bonding brake linings and clutch facings to metals, in braking applications involving high frictional shearing forces.

6.2 Ordering data. Procurement documents should specify the following:

a) Title, number and date of this specification.

b) Title, number and date of applicable documents.

c) Form of adhesive required (see 1.2)

1) If liquid; capacity and kind of container and quantity required (see 5.1.1.1).

2) If film; thickness of film, width and length of rolls and number of rolls required (see 3.1.2).

d) Amount of solvent if required (see 3.1.3).

e) Number of instruction sheets required (see 3.4).

f) Applicable levels of packaging and packing required (see Section 5).

g) When a mutual safety data sheet is required (see 3.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

6.3 Any question raised regarding toxicity should be referred to appropriate medical authority. In the case of Army procurement, the Surgeon General of the Army will act as advisor to the procuring activity.

6.4 Brake fluid. Information as to manufacturers of brake fluids to meet VV-B-680, MIL-B-46176 may be obtained by applying to Commander, U.S. Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060.

Custodians:

Army - MR
Navy - YD
Air Force - 99

Preparing Activity:

Army - MR

Project No. 8040-0348

Review activities:

Army - AR, AL

User activities:

Army - MI, ME

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-46091B Adhesive, Brake Lining to Metal

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:49Z
Check the source to verify that this is the current version before use.